Gabriel Rabanal

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WORK EXPERIENCE

4G Clinical — Wellesley, MA
Software Engineer — Sep 2024 — Present

- Developed Python automation and reusable validation scripts to implement complex study logic, prevent configuration errors, and ensure reproducible configurations
- Developed and maintained production configurations for clinical study platforms, ensuring validated data workflows and reporting in a regulated (GxP) environment
- Collaborated with engineering and clients to troubleshoot issues and ensure reliable system behavior and audit-ready configurations

Harvard University — Cambridge, MA / Geneva, Switzerland
Doctoral Researcher (ATLAS/CERN; ML & large-scale data analysis) — Sep 2018 — Nov 2023

- Developed and validated ML classifiers (DNNs, BDTs) on multi-terabyte particle physics datasets using HPC batch computing; improved signal detection significance by +3σ
- Built automated data-quality and model-validation pipelines for large-scale analyses to reduce performance regressions and improve reliability (statistical tests, reproducibility checks, uncertainty quantification)
- Commissioned and calibrated detector subsystems to improve tracking efficiency and spatial resolution; mentored researchers and coordinated cross-institution deliverables

Yale University — New Haven, CT
Lab Researcher — Jan 2016 — Apr 2016

- Analyzed detector data and built Monte Carlo simulations for the PROSPECT neutrino experiment

Peruvian Institute of Nuclear Energy — Lima, Peru
Undergraduate Researcher — Jan 2015 — Apr 2015

- Calibrated spectrometers for neutron flux measurements in a research nuclear reactor supporting medical isotope production

EDUCATION

Harvard University — Cambridge, MA
Ph.D. in Physics — Nov 2023

National University of Engineering — Lima, Peru
B.S. in Physics, GPA: 4.00/4.00 — Jan 2016

SKILLS

Programming: Python (advanced), C/C++ (advanced), SQL (working knowledge)
Machine Learning: TensorFlow/Keras, XGBoost, scikit-learn; feature engineering, model validation
Data & Statistics: NumPy, Pandas, SciPy; Bayesian and frequentist methods; data pipelines
Tools: Git, Linux/Bash, Jupyter, VS Code; testing, debugging, HPC environments
Languages: Spanish (native), English (fluent), French (advanced)
Other: Linguistics; Akkadian, Sumerian, Quechua, Latin

SELECTED PUBLICATIONS

ATLAS Collaboration. Observation of VVZ production at $\sqrt{s} = 13$ TeV (Phys. Lett. B, 2025) [arXiv]
ATLAS Collaboration. Evidence for three massive vector bosons (Phys. Lett. B, 2019) [arXiv]
Rabanal Bolaños, G. Micromegas sectors for the ATLAS muon upgrade (PoS ICHEP2020, 2021)